

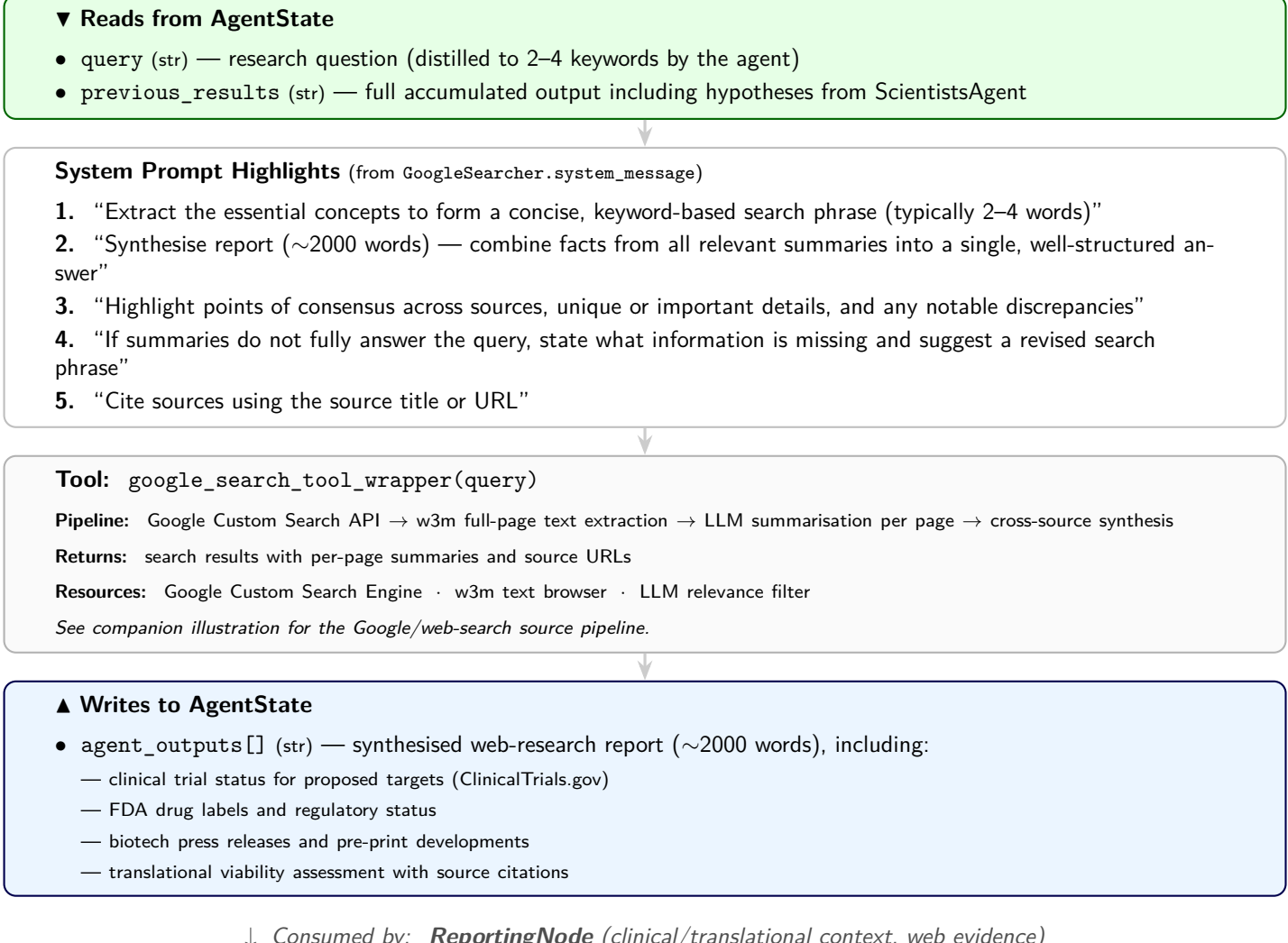


Figure 4: **GoogleSearcher** — workflow interface view. This agent is typically invoked late in the pipeline—often after ScientistsAgent has produced mechanistic hypotheses—but the planner may invoke it at any point, omit it for purely molecular queries, or re-invoke it after a replan cycle. Its system prompt defines a disciplined three-step workflow: (1) distil the query into 2–4 core keywords, (2) execute a targeted Google Custom Search, and (3) synthesise the retrieved page summaries into a cohesive ~2000-word report. Unlike the PubMed agent, its value lies in surfacing information *not* indexed in biomedical databases—clinical trial registries, FDA labels, biotech press releases, and pre-prints—thereby grounding the proposed targets in translational reality (see companion illustration for the web-search pipeline).

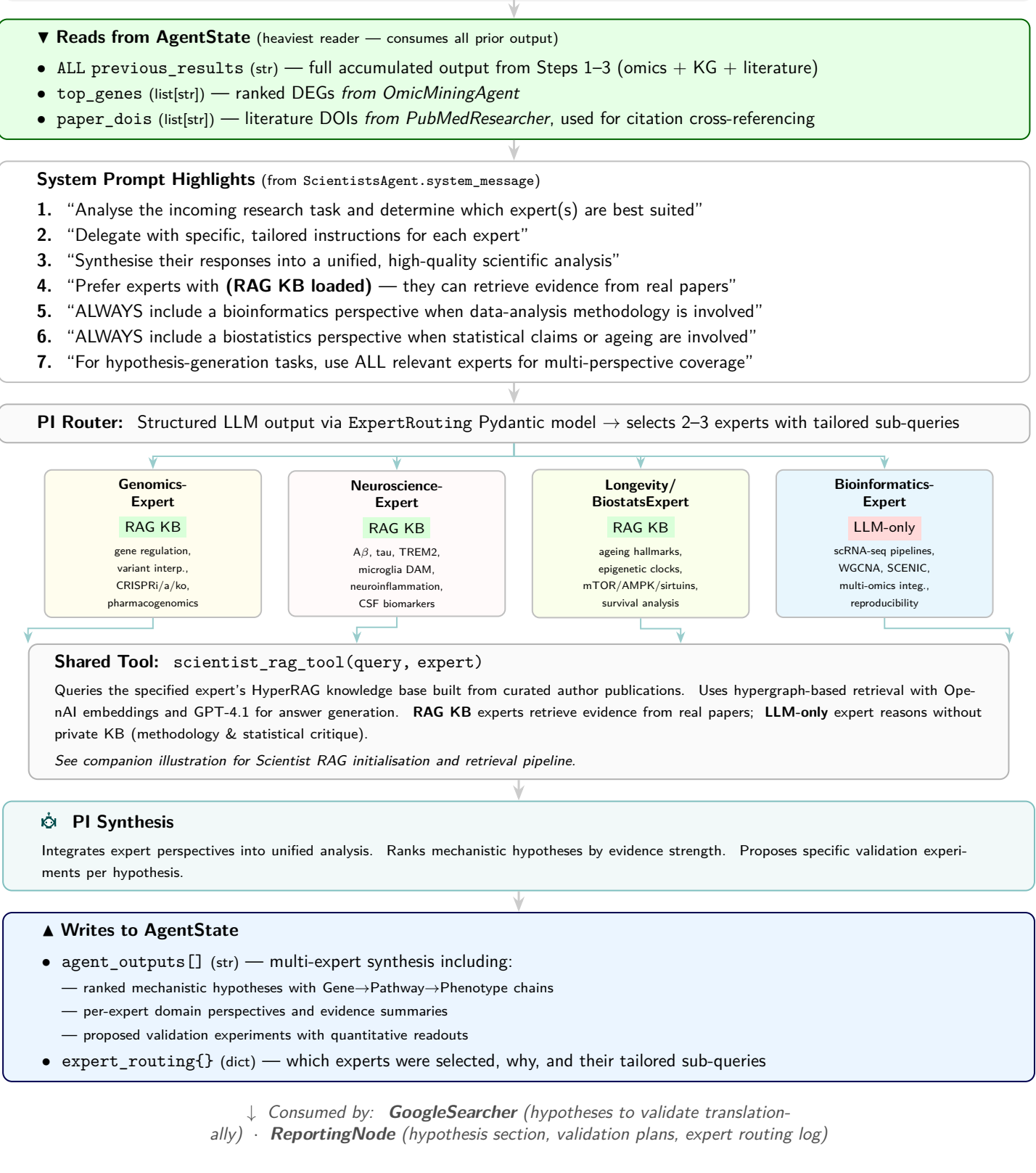


Figure 5: **ScientistsAgent** — workflow interface view. In the example realisation shown here this compound agent runs fourth, but the LLM planner may invoke it earlier, later, or re-invoke it after a replan cycle—for instance, if literature search reveals novel targets that warrant fresh expert synthesis. It is the heaviest reader of **AgentState**, consuming all prior omics, KG, and literature output. A PI-level router uses a structured **ExpertRouting** Pydantic model to select 2–3 domain experts and dispatch tailored sub-queries. The system prompt distinguishes **RAG KB loaded** experts (Genomics, Neuroscience, Longevity/Biostatistics), which retrieve evidence from curated author publications via HyperRAG, from the **LLM-only** BioinformaticsExpert, preferred for methodology and statistical critique. Routing rules mandate a bioinformatics perspective for data-analysis tasks, a biostatistics perspective for statistical claims, and all relevant experts for hypothesis generation (see companion illustration for RAG initialisation and retrieval pipeline). The PI synthesis node integrates expert perspectives into ranked hypotheses with proposed validation experiments.



Figure 6: **Author Knowledge-Base Build Pipeline**. A gitignored **experts.yaml** maps each expert alias to a private scientist name used *only* for PubMed queries. The offline build (Stages 1–6) fetches papers, extracts text, chunks, embeds, and runs LLM entity/relation extraction. At query-time the pre-built KB loads from cache in seconds. This separation provides (i) *modularity* (experts updated independently), (ii) *efficiency* (build once, query many), and (iii) *privacy* (real names never reach the agent). See companion illustration for the HyperRAG retrieval architecture at query-time.